

Unit 12, Lesson 4: Factoring Expressions

Benchmark

MA.8.AR.1.3 Rewrite the sum of two algebraic expressions having a common monomial factor as a common factor multiplied by the sum of two algebraic expressions.

Learning Target

- ☐ I can rewrite an algebraic expression with a common factor using the distributive property.

12.4.1 Warm-Up: Number Puzzle

1. Solve the puzzle by filling in each blank cell. The values in the blue box represent the products of the values at the end of each row and column.

49		
21	36	

12.4.2 Refresher: Common Factors

In multiplication expressions, the numbers being multiplied are called factors and the result is called the product.

Numbers can have common factors.



A **common factor** of two numbers is a factor that they both share.

For example, 1, 3, 5, 9, 15 and 45 are factors of 45, and 1, 2, 3, 4, 5, 6, 10, 12, 15, 20, 30, and 60 are factors of 60.

The common factors 45 and 60 are 1, 3, 5, and 15.



The **greatest common factor** (GCF) of two or more whole numbers is the largest whole number that evenly divides the given whole numbers.

For example, 15 is the greatest common factor for 45 and 60.

1. For the number pairs in the table, list the common factors. Then, identify the GCF.

12 and 42	10 and 35
Common factors:	Common factors:
GCF = _____	GCF = _____

Common factors can be used to rewrite numeric expressions.

2. Consider the expression $24 + 32$.

a. List all common factors of 24 and 32.

b. Complete the steps to rewrite the sum of 24 and 32 using their GCF.

$$24 + 32$$

$$(8 \times \underline{\quad}) + (8 \times \underline{\quad})$$

$$8(\underline{\quad} + \underline{\quad})$$

c. Choose another common factor of 24 and 32 to write a different equivalent expression to $24 + 32$.

$$24 + 32$$

$$(\underline{\quad} \times \underline{\quad}) + (\underline{\quad} \times \underline{\quad})$$

$$\underline{\quad}(\underline{\quad} + \underline{\quad})$$

12.4.3 Activity: Card Sort

Six numeric expressions are shown. Work with your partner to find at least one equivalent expression for each. Expressions may have one, two, or three matches.

Record the letter(s) of the equivalent expressions in the space provided under each expression.

1. $90 - 40$	2. $18 + 66$	3. $20 - 80$
4. $-30 + 45$	5. $49 + 21$	6. $72 + 54$

12.4.4 Exploration: Equivalent Forms of Algebraic Expressions

Common factors can also be used to rewrite algebraic expressions.

1. Use your understanding of common factors to complete the area models shown.

5 $-2y$		a -6		 	
-3	-15 $6y$		5	 	
 		 $-x$		 	
-4	$-16x$ 4			30 $-6x$	
 		 		 	
 		 		$-8x$ 28	

An area model can be used to show the relationship between equivalent forms of algebraic expressions.

In the area models in the previous problem, the factored form of an expression is the *width* $\times$ *length* of the larger rectangle. An equivalent form can be represented by the sum of the areas of each smaller rectangle.

2. Complete the table using the area models from the previous problem.

Factored Form	Equivalent Expression
$-3(5 - 2y)$	$-15 + 6y$
$5(a - 6)$	
	$6a - 2b$
	$-16z + 4$
	$3d - 6x$
	$-8x + 28$

3. Compare your work with your partner. Discuss any differences and make any corrections as needed.

4. Work with your partner to complete the following.

a. Without using area models, complete the table.

Factored Form	Equivalent Expression
$-4(2w - 5x)$	
$-(2x - 3y)$	
	$20x - 10y$
$4(a - 11)$	
	$10a - 25b$
$-2(3y - z)$	

b. Explain how you determined the factored form of the expression $10a - 25b$.

5. For the expression $20x - 10y$, Pao and Camiya wrote different factored forms of the expression.

Pao's Expression	Camiya's Expression
$5(4x - 2y)$	$10(2x - y)$

- a. Explain who wrote a correct equivalent expression in factored form.
- b. What is another equivalent expression using a different common factor?
- c. Which expression is equivalent to $20x - 10y$, using the GCF of $20x$ and $-10y$?

12.4.5 Bring It Together: Factoring Expressions

Properties of operations can be used in different ways to create equivalent expressions.

The distributive property can be used to expand an expression. For example, the expression $3(x + 5)$ can be rewritten as $3(x) + 3(5)$ or $3x + 15$.

The distributive property can also be used to factor an expression. For example, the expression $8x + 12$ can be rewritten as $2(4x + 6)$ or $4(2x + 3)$.

Equivalent expressions can also be generated using the commutative property of addition or multiplication.

$$(4x + 6)2 \qquad 4(3 + 2x)$$

1. Complete the following to factor the expression $12d - 36$.

- a. Choose a common factor of the terms $12d$ and -36 .

☐ 2 ☐ 3 ☐ 4 ☐ 6 ☐ 12

- b. What is the result of dividing $12d$ by the common factor selected?

- c. What is the result of dividing -36 by the common factor selected?

- d. Use your answers to Parts A-C to write an expression equivalent to $12d - 36$ in factored form.

$$12d - 36 = \underline{\hspace{1cm}}(\underline{\hspace{1cm}} + \underline{\hspace{1cm}})$$

- e. Choose a different common factor from Part A, and use it to write a different equivalent expression.

$$12d - 36 = \underline{\hspace{1cm}}(\underline{\hspace{1cm}} + \underline{\hspace{1cm}})$$

2. Several expressions are shown.

$8(3 + m)$	$2(8m + 12)$	$4(2m + 6)$
$(2m + 8)3$	$(1 + 3)8m$	$(12 + 4m)2$

- a. Circle the expressions equivalent to $8m + 24$.
- b. Put an asterisk (*) next to the equivalent expression that has the GCF of $8m$ and 24 as one of its factors.
- c. Put a smiley face (☺) next to an expression where more than one property of operations can be used to show it is equivalent to $8m + 24$.

12.4.6 Try It: Factoring Expressions

1. Rewrite each expression in an equivalent factored form.

a. $-15x + 20 =$

b. $4x - 32 =$

c. $-27 - 12d =$

d. $35f + 28g =$

2. Fill in the blanks of the equation to make it true.

$$75a + 25b = \underline{\hspace{1cm}}(\underline{\hspace{1cm}}a + b)$$

12.4.7 Wrap-Up: Cool Down

Janelle missed math class today. She understands using the distributive property to expand an expression, but she doesn't know how to rewrite the sum of two terms with a common factor in factored form.

1. Explain to Janelle how to rewrite an expression like $36a - 16$ in an equivalent factored form.

